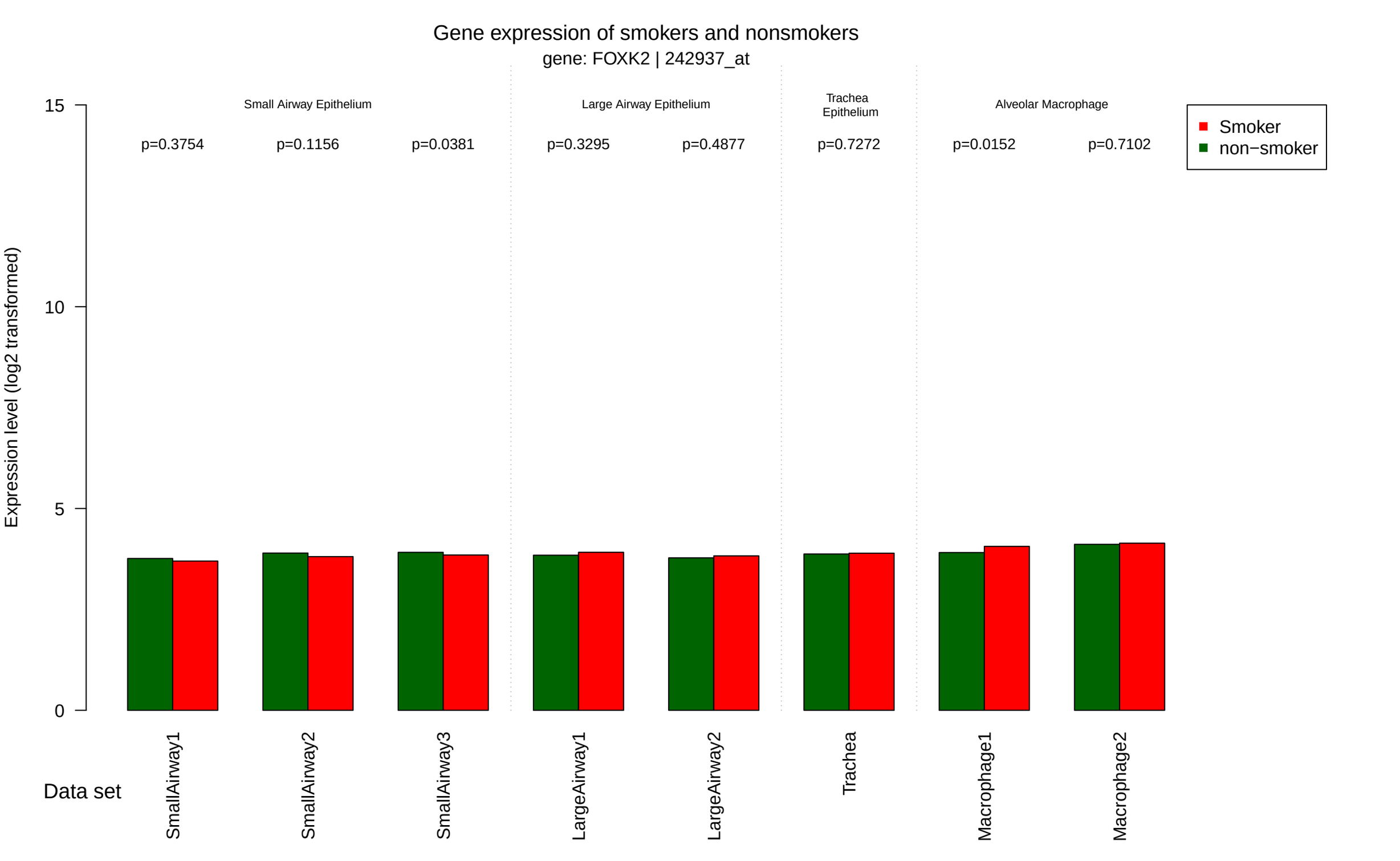
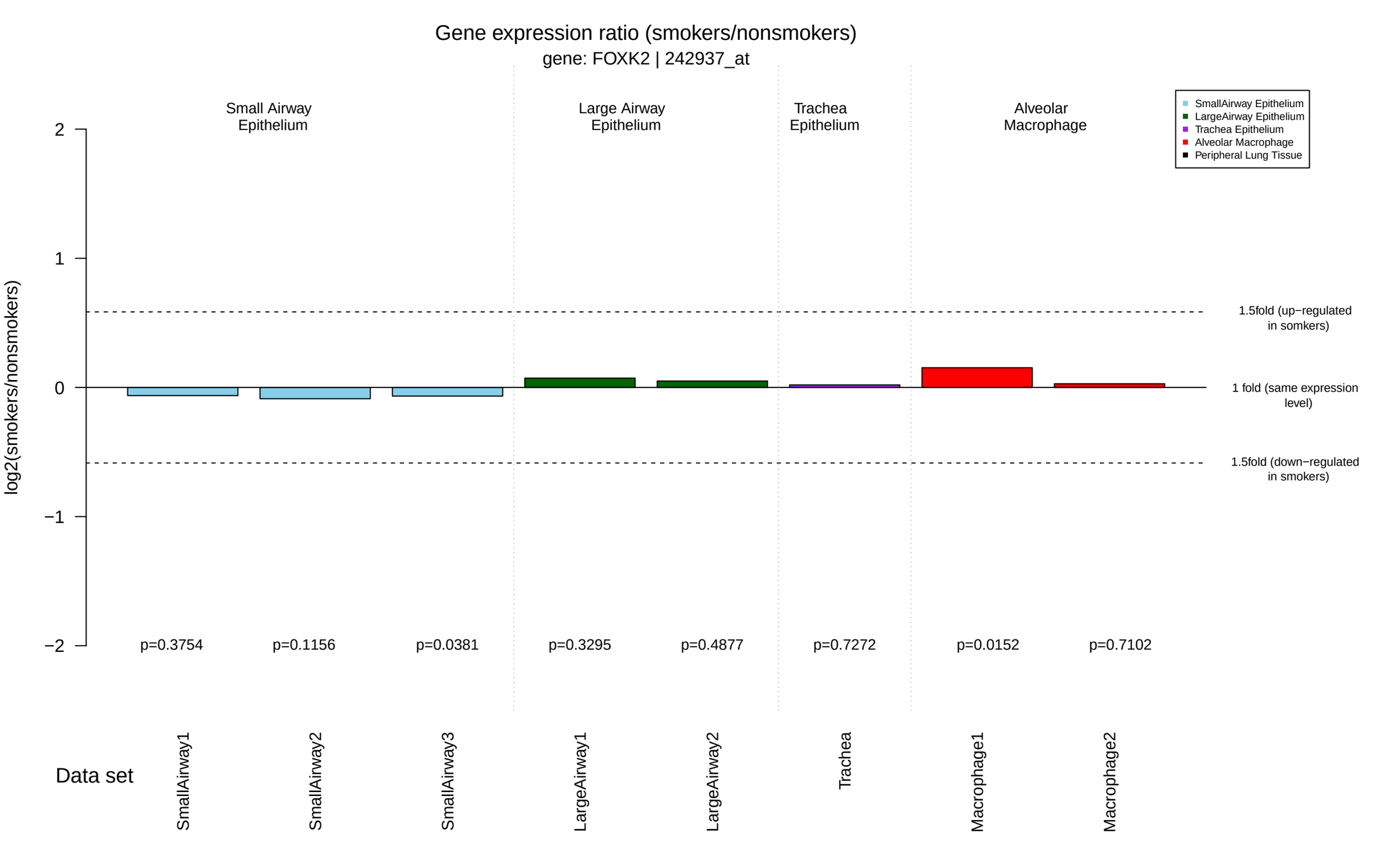
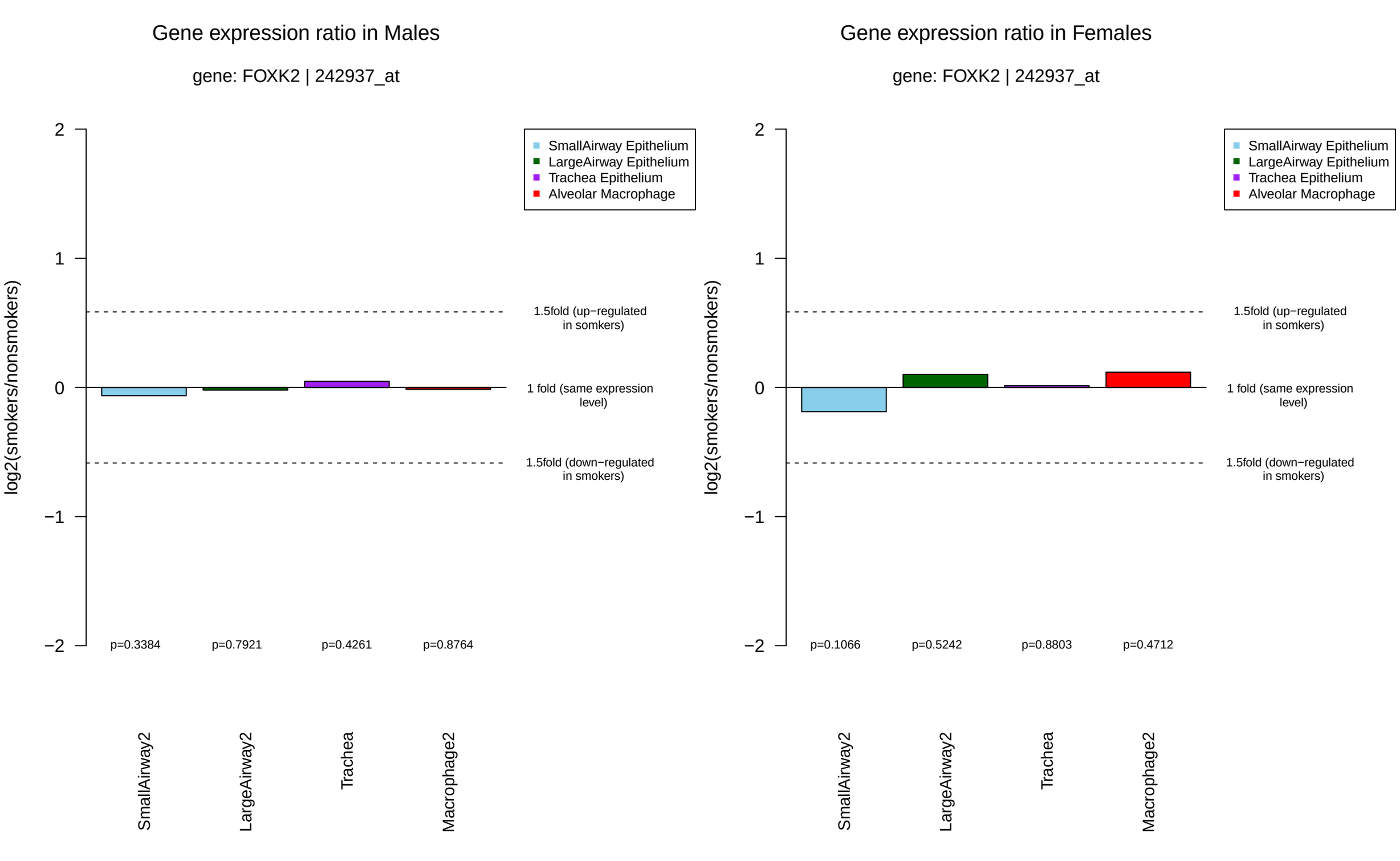
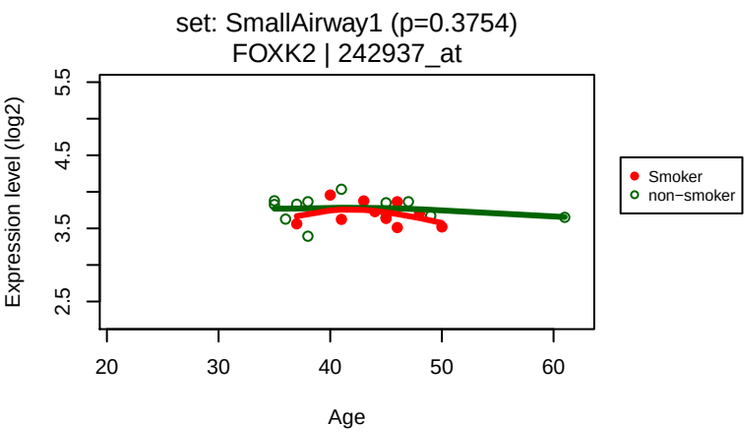
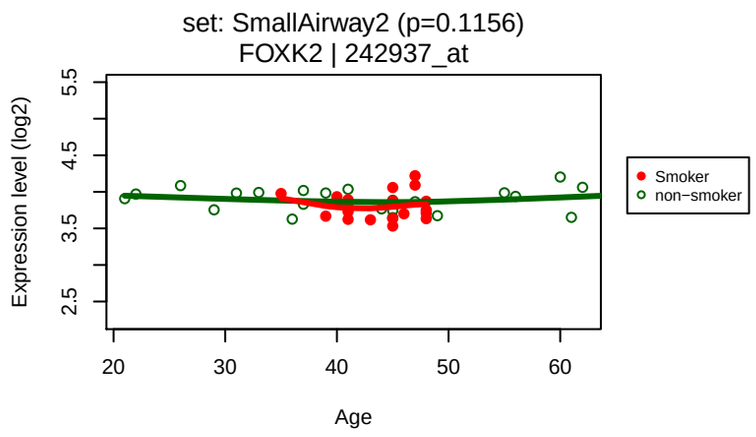
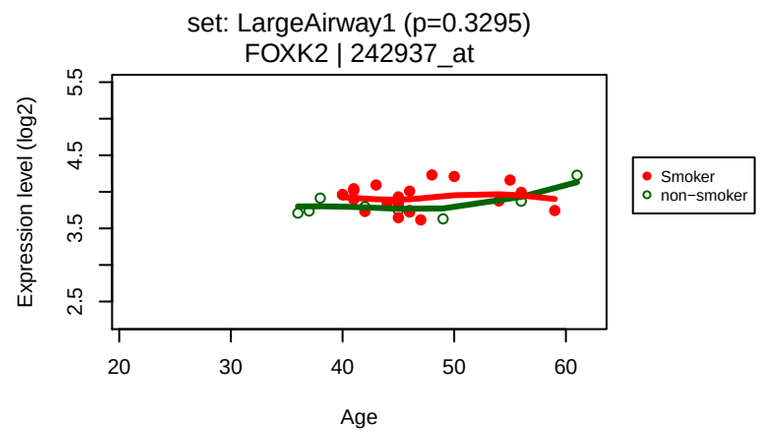

Gene expression of smokers and nonsmokers
gene: FOXK2 | 242937_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3754
p=0.1156
p=0.0381
p=0.3295
p=0.4877
p=0.7272
p=0.0152
p=0.7102
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

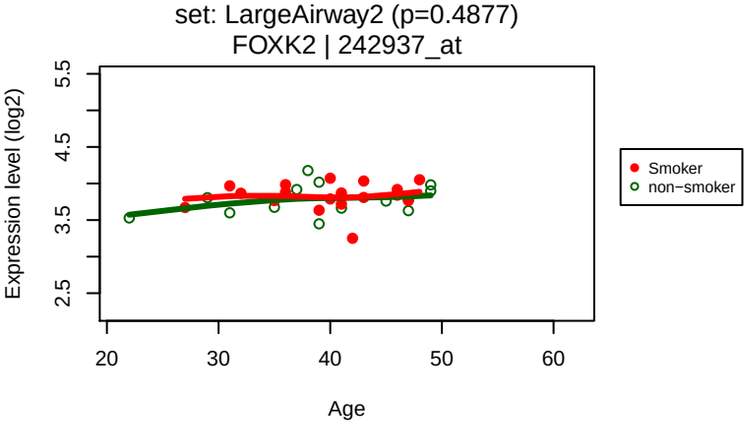

Gene expression ratio (smokers/nonsmokers)
gene: FOXK2 | 242937_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3754
p=0.1156
p=0.0381
p=0.3295
p=0.4877
p=0.7272
p=0.0152
p=0.7102
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

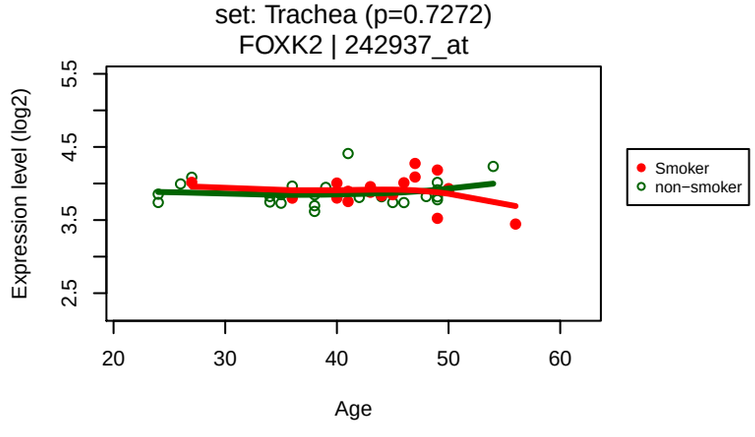

Gene expression ratio in Males
gene: FOXK2 | 242937_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3384
p=0.7921
p=0.4261
p=0.8764
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FOXK2 | 242937_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1066
p=0.5242
p=0.8803
p=0.4712
SmallAirway2
LargeAirway2
Trachea
Macrophage2

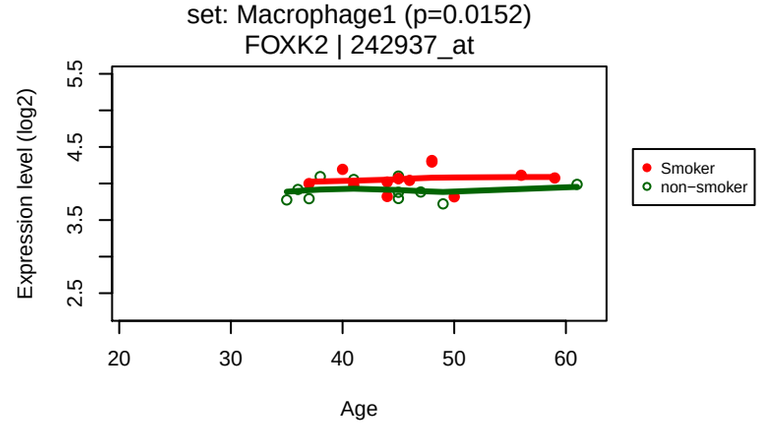

set: SmallAirway1 (p=0.3754)
FOXK2 | 242937_at
Expression level (log2)
Age
Smoker
non−smoker

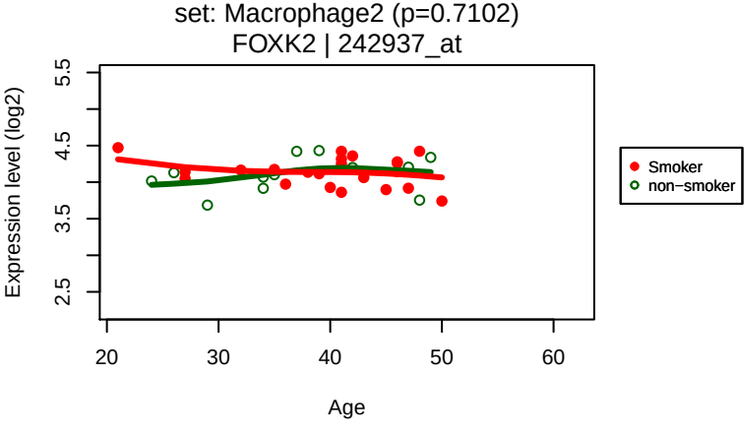

set: SmallAirway2 (p=0.1156)
FOXK2 | 242937_at
Expression level (log2)
Age
Smoker
non−smoker

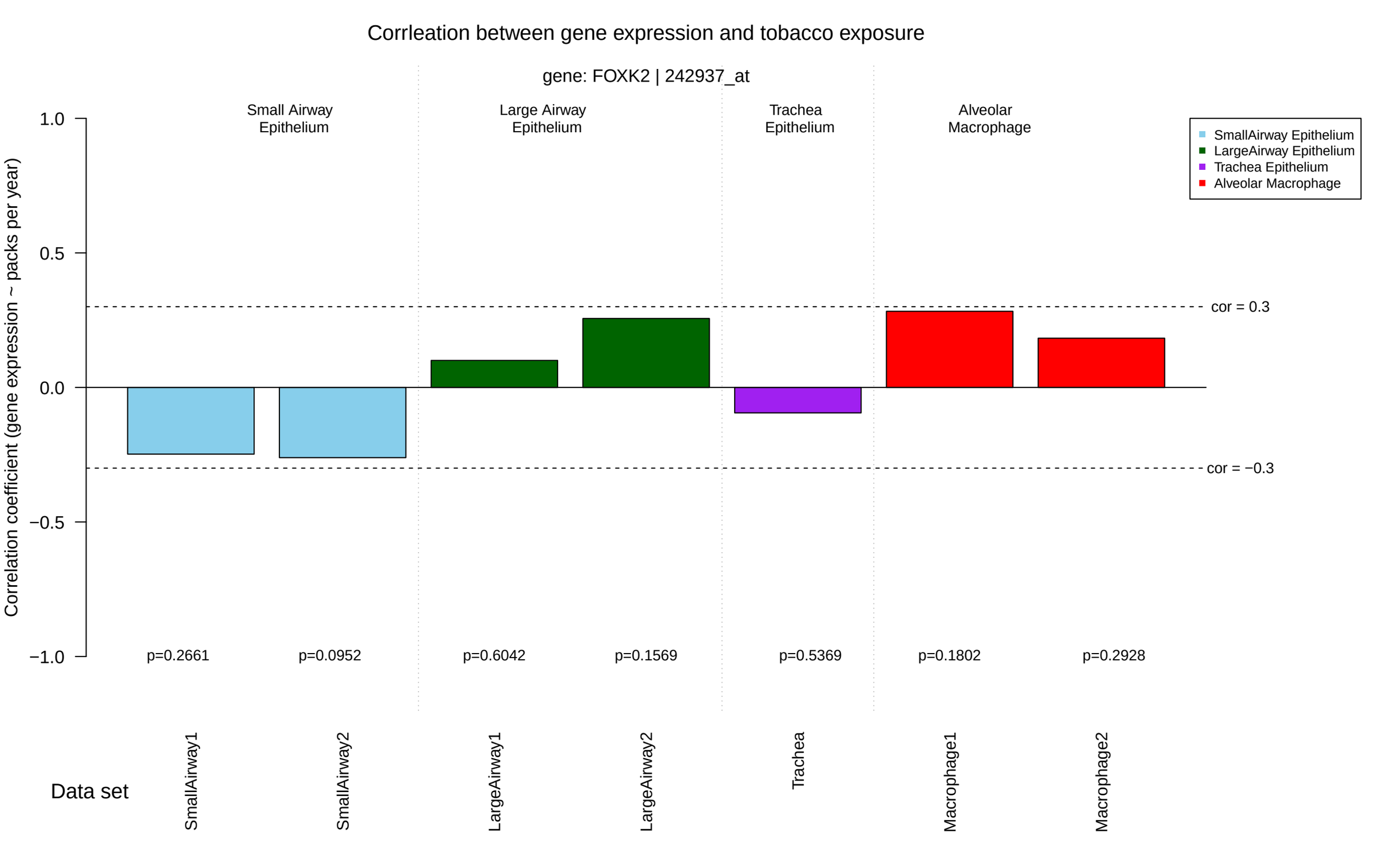

set: LargeAirway1 (p=0.3295)
FOXK2 | 242937_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4877)
FOXK2 | 242937_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7272)
FOXK2 | 242937_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0152)
FOXK2 | 242937_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7102)
FOXK2 | 242937_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FOXK2 | 242937_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2661
p=0.0952
p=0.6042
p=0.1569
p=0.5369
p=0.1802
p=0.2928
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2